

DATABASE SYSTEMS

CHAP. 4: Relational Algebra

Relational Algebra

- The basic set of operations for the relational model.
- Enable a user to specify basic retrieval requests.

Relational Algebra

- The result of a retrieval is a new relation, which may have been formed from one or more relations.
- The algebra operations thus produce new relations, which can be further manipulated using operations of the same algebra.

Relational Algebra

- A sequence of relational algebra operations forms a relational algebra expression, whose result will also be a relation that represents the result of a database query (or retrieval request).
- Relational Algebra is a Procedural Paradigm: You Need to Tell What/How to Construct the Result

Basic Relational Operations

- **Unary Operations**
 - SELECT: σ
 - PROJECT: π
 - Rename operator: ρ

Basic Relational Operations

Binary Operations

Set operations:

UNION: $\cup$

INTERSECTION: $\cap$

DIFFERENCE : -

DIVISION: $\div$

CARTESIAN PRODUCT: $\times$

JOIN operations: 

SELECT

- Select a subset of the tuples from a relation that satisfy a selection condition.
- It is a filter that keeps only those tuples that satisfy a qualifying condition – those satisfying the condition are selected while others are discarded.

SELECT

□ Relation r

<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
α	α	1	7
α	β	5	7
β	β	12	3
β	β	23	10

■ $\sigma_{A=B \wedge D > 5}(r)$

<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
α	α	1	7
β	β	23	10

PROJECT

- Selects certain columns from the table and discards the other columns.
- Creates a vertical partitioning – one with the needed columns containing results of the operation and other containing the discarded columns.

PROJECT

Relation r :

A	B	C
α	10	1
α	20	1
β	30	1
β	40	2

$\Pi_{A,C}(r)$

A	C
α	1
α	1
β	1
β	2

=

A	C
α	1
β	1
β	2

RENAME

- We may want to apply several relational algebra operations one after the other.
- Either we can write the operations as a single relational algebra expression by nesting the operations, or we can apply one operation at a time and create intermediate result relations.
- In the latter case, we must give names to the relations that hold the intermediate results.

RENAME

We may *rename* the attributes of a relation or the relation name or both

- $\rho_{S(B_1, B_2, \dots, B_n)}(R)$
 - ◆ Change the relation name from R to S , *and*
 - ◆ Change column (attribute) names to $B_1, B_2, \dots, B_n$
- $\rho_S(R)$:
 - ◆ Change the *relation name* only to S
- $\rho_{(B_1, B_2, \dots, B_n)}(R)$:
 - ◆ Change the *column (attribute) names* only to $B_1, B_2, \dots, B_n$

RENAME

- | Change relation name to Emp and attributes to First_name, Last_name, and Salary

$\rho_{\text{Emp}(\text{First_name}, \text{Last_name}, \text{Salary})} (\pi_{\text{FNAME}, \text{LNAME}, \text{SALARY}} (\text{EMPLOYEE}))$

or

$\text{Emp}(\text{First_name}, \text{Last_name}, \text{Salary}) \leftarrow \pi_{\text{FNAME}, \text{LNAME}, \text{SALARY}} (\text{EMPLOYEE})$

UNION

- Result is a relation that includes all tuples that are either in R or in S or in both R and S.
- Duplicate tuples are eliminated.

UNION

Relations r , s :

A	B
α	1
α	2
β	1

r

A	B
α	2
β	3

s

$r \cup s$:

A	B
α	1
α	2
β	1
β	3

INTERSECTION

- Relation that includes all tuples that are in both R and S.
- The two operands must be "type compatible"

INTERSECTION

□ Relation r, s :

A	B
α	1
α	2
β	1

r

A	B
α	2
β	3

s

□ $r \cap s$

A	B
α	2

DIFFERENCE

- Relation that includes all tuples that are in R but not in S .
- The two operands must be "type compatible".

DIFFERENCE

Relations r, s :

A	B
α	1
α	2
β	1

r

A	B
α	2
β	3

s

$r - s$:

A	B
α	1
β	1

CARTESIAN PRODUCT

- Concatenation of every tuple of one relation with every tuple of a second relations.
- Cartesian product of relation a (having m tuples) and relation b (having n tuples) has m times n tuples.
- Cartesian product is denoted $a \times b$ or a times b .

CARTESIAN PRODUCT

Relations r, s :

A	B
α	1
β	2

r

C	D	E
α	10	a
β	10	a
β	20	b
γ	10	b

s

$r \times s$:

A	B	C	D	E
α	1	α	10	a
α	1	β	10	a
α	1	β	20	b
α	1	γ	10	b
β	2	α	10	a
β	2	β	10	a
β	2	β	20	b
β	2	γ	10	b

JOIN OPERATIONS

- Join operation is essentially a cartesian product followed by a selection criterion.
- Join operation denoted by $\bowtie$.
- JOIN operation also allows joining variously related tuples from different relations.

Types of JOIN

- Inner Joins:
 - Theta join
 - EQUI join
 - Natural join
- Outer join:
 - Left Outer Join
 - Right Outer Join
 - Full Outer Join

Inner JOIN

- In an inner join, only those tuples that satisfy the matching criteria are included, while the rest are excluded.

Outer JOIN

- In an outer join, along with tuples that satisfy the matching criteria, we also include some or all tuples that do not match the criteria.

JOIN OPERATIONS

Relations r, s :

A	B	C	D
α	1	α	a
β	2	γ	a
γ	4	β	b
α	1	γ	a
δ	2	β	b

r

B	D	E
1	a	α
3	a	β
1	a	γ
2	b	δ
3	b	ϵ

s

$r \bowtie s$

A	B	C	D	E
α	1	α	a	α
α	1	α	a	γ
α	1	γ	a	α
α	1	γ	a	γ
δ	2	β	b	δ

DIVISION

- The division operator is used for queries which involve the 'all'.
- $R1 \div R2$ = tuples of R1 associated with all tuples of R2.

DIVISION

R

ENO	PNO	PNAME	BUDGET
E1	P1	Instrumentation	150000
E2	P1	Instrumentation	150000
E2	P2	Database Develop.	135000
E3	P1	Instrumentation	150000
E3	P4	Maintenance	310000
E4	P2	Instrumentation	150000
E5	P2	Instrumentation	150000
E6	P4	Maintenance	310000
E7	P3	CAD/CAM	250000
E8	P3	CAD/CAM	250000

*Find the
employees who
work for both
project P1 and
project P4?*

S

PNO	PNAME	BUDGET
P1	Instrumentation	150000
P4	Maintenance	310000

R ÷ S

ENO
E3